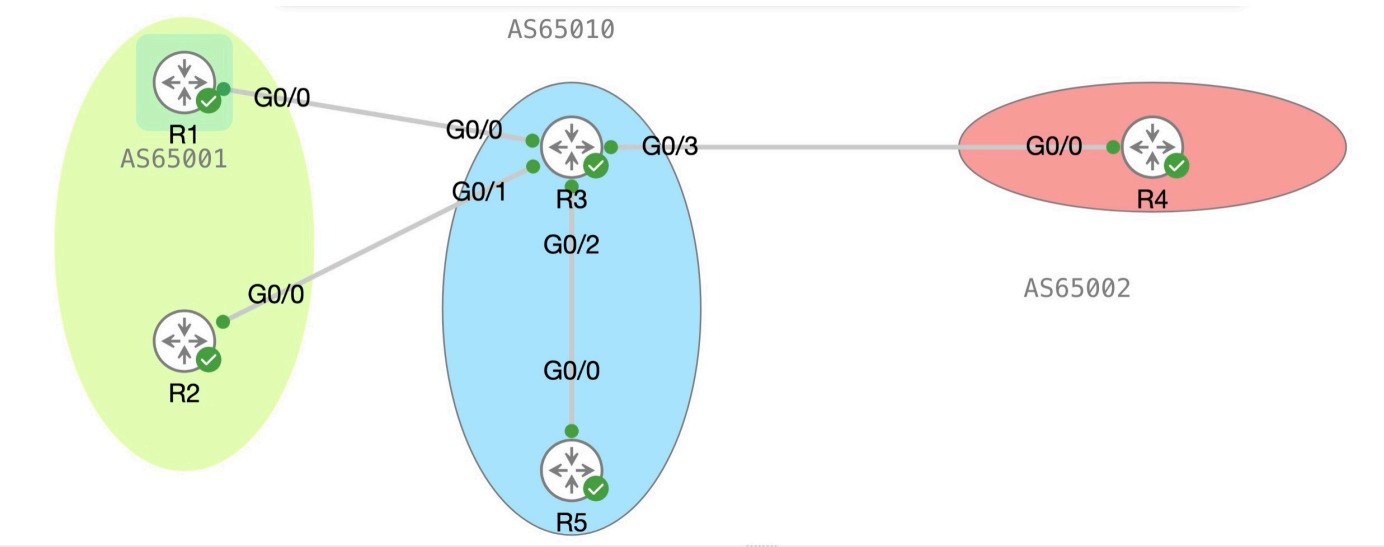


Network topology



IP addressing plan

Router Name	Role	Autonomous System #
R1	ISP-A edge 1	65001
R2	ISP-A edge 2	65001
R3	Enterprise edge	65010
R4	ISP-B	65002
R5	Enterprise internal	65010

Interface IP mapping

Between R1 and R3

10.0.13.0/30

Between R2 and R3

10.0.23.0/30

Between R4 and R3

10.0.34.0/30

Between R3 and R5

10.0.35.0/30

Advertised networks

Enterprise:

192.168.100.0/24

ISP-A:

203.0.113.0/24

ISP-B:

198.51.100.0/24

Configuring the routers

- On R1, ISP-A

```
en
conf t
hostname R1

int gi0/0
ip address 10.0.13.1 255.255.255.252
no sh
exit

int Loopback0
ip address 203.0.113.1 255.255.255.0
exit
```

- On R2, ISP-A

```
en
conf t
hostname R2

int GigabitEthernet0/0
ip add 10.0.23.1 255.255.255.252
no sh
exit

int Loopback0
ip add 203.0.114.1 255.255.255.0
exit
```

- On R4, ISP-B

```
en
conf t
hostname R4

int GigabitEthernet0/0
ip add 10.0.34.1 255.255.255.252
no shutdown
exit

int Loopback0
ip add 198.51.100.1 255.255.255.0
exit
```

- On R5, Internal Router

```
en
conf t
hostname R5

int gi0/0
ip add 10.0.35.2 255.255.255.252
no shutdown
exit

int Loopback0
ip add 192.168.200.1 255.255.255.0
exit
```

- On R3, Edge Router

```
en
conf t
hostname R3

int gi0/0
ip add 10.0.13.2 255.255.255.252
no sh
exit

int gi0/1
ip add 10.0.23.2 255.255.255.252
```

```
no sh
exit

int gi0/2
ip add 10.0.34.2 255.255.255.252
no sh
exit

int gi0/3
ip add 10.0.35.1 255.255.255.252
no sh
exit

int Loopback0
ip add 192.168.100.1 255.255.255.0
exit
```

Configuring the BGP protocol

- On R1, ISP-A

```
router bgp 65001
  neighbor 10.0.13.2 remote-as 65010
  network 203.0.113.0 mask 255.255.255.0
exit
```

- On R2, ISP-A

```
router bgp 65001
  neighbor 10.0.23.2 remote-as 65010
  network 203.0.114.0 mask 255.255.255.0
exit
```

- On R4, ISP-B

```
router bgp 65002
  neighbor 10.0.34.2 remote-as 65010
  network 198.51.100.0 mask 255.255.255.0
exit
```

- On R5, Internal Router

```
router bgp 65010
  neighbor 10.0.35.1 remote-as 65010
  network 192.168.200.0 mask 255.255.255.0
exit
```

- On R3, Edge Router

```
router bgp 65010
  neighbor 10.0.13.1 remote-as 65001
  neighbor 10.0.23.1 remote-as 65001
  neighbor 10.0.34.1 remote-as 65002
  neighbor 10.0.35.2 remote-as 65010
  network 192.168.100.0 mask 255.255.255.0
exit
```

Multi-exit discriminator (MED) configuration

We will have ISP-A advertise the same prefix through R1 and R2. Company wants to control which ISP entrance is preferred.

Add the same loopback network to R2:

```
interface Loopback1
  ip address 203.0.113.2 255.255.255.0
exit

router bgp 65001
  network 203.0.113.0 mask 255.255.255.0
```

Then apply MED on R2:

```
route-map HIGH-MED permit 10
  set metric 200
exit

router bgp 65001
  neighbor 10.0.23.2 route-map HIGH-MED out
exit
```

R1 has the default MED of 0. Therefore:

```
R1 → 203.0.113.0/24 → MED 0
R2 → 203.0.113.0/24 → MED 200
```

R3 should prefer the R1 path.

AS-path prepending

Company wants ISP-A to be preferred inbound path

On R3:

```
route-map ISP-B-PREPEND permit 10
  set as-path prepend 65010 65010 65010
exit

router bgp 65010
  neighbor 10.0.34.1 route-map ISP-B-PREPEND out
exit
```

ISP-B now sees:

ISP-A path:

65010

ISP-B path:

65010 65010 65010 65010

The ISP-B path appears less attractive.

BGP Communities

On R3, we will apply the BGP tags so when the ISPs see the traffic, they see the tag for their own decision making (policy decisions)

```
route-map ENTERPRISE-COMMUNITY permit 10
  set community 65010:100
exit

router bgp 65010
  neighbor 10.0.13.1 send-community
  neighbor 10.0.13.1 route-map ENTERPRISE-COMMUNITY out
  neighbor 10.0.23.1 send-community
  neighbor 10.0.23.1 route-map ENTERPRISE-COMMUNITY out
  neighbor 10.0.34.1 send-community
  neighbor 10.0.34.1 route-map ENTERPRISE-COMMUNITY out
exit
```